

A REPORT

*submitted in partial fulfilment of the requirements
of the course CS517 - Digital Image Processing*

Image Matching Challenge

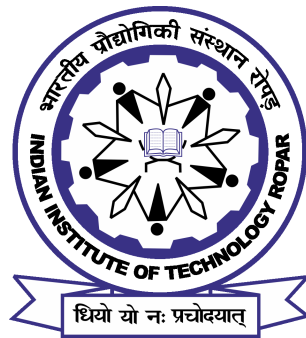
submitted by

Vaibhav Singla (2021MCB1254)
Jai Anurag Y (2022CSB1086)

Students of B.Tech, Mathematics and Computing, Computer Science and Engineering

under the guidance of

Dr. Puneet Goyal



DEPARTMENT OF COMPUTER SCIENCE
INDIAN INSTITUTE OF TECHNOLOGY ROPAR
RUPNAGAR - 140001
PUNJAB - INDIA.

May 2025

Abstract

This report presents a solution to the 2025 Image Matching Challenge, which focuses on two critical tasks in computer vision: grouping visually similar images into coherent scenes and estimating camera poses to enable 3D scene reconstruction. The challenge mimics real-world complexities by introducing unordered, noisy, and potentially unrelated image sets. Our approach leverages ORB (Oriented FAST and Rotated BRIEF) for efficient keypoint detection and binary descriptor extraction, followed by Brute-Force matching with Hamming distance to identify visual correspondences between image pairs. Based on the strength of these matches, we apply clustering techniques to group images into scenes and identify outliers. The methodology aligns with the challenge’s dual evaluation criteria—mean Average Accuracy (mAA) and clustering score—by ensuring accurate scene reconstruction and robust image partitioning. The report details the preprocessing steps, feature matching pipeline, and rationale for the use of key inbuilt functions, demonstrating a practical and scalable solution for large-scale image matching and 3D reconstruction tasks.

Contents

1	Introduction	4
1.1	Overview of the Challenge	4
1.2	Objective of the Report	4
2	Problem Description	5
2.1	Challenge Description	5
2.2	Evaluation Metrics	5
3	Dataset Overview	7
3.1	Dataset Structure	7
3.2	Data Preprocessing	8
4	Methodology	9
4.1	Approach to the Challenge	9
4.2	Inbuilt Functions Used	10
5	Results	13
6	Discussion	16
6.1	Insights from the Experiment	16
6.2	Limitations and Potential Improvements	17
7	Conclusion and Future Work	19

Chapter 1

Introduction

1.1 Overview of the Challenge

The Image Matching Challenge is a computer vision competition focused on advancing methods for 3D scene reconstruction from unordered and potentially unrelated image collections. In its third edition, the 2025 challenge builds upon the 2024 Image Matching Challenge by introducing an additional layer of complexity: participants must not only reconstruct 3D scenes but also determine how images should be grouped into coherent scenes or discarded as outliers.

This problem mimics real-world scenarios where online image collections are often noisy, unorganized, and contain visually similar or entirely unrelated images. Traditional Structure-from-Motion (SfM) pipelines, while effective in controlled environments, struggle with such diverse data. The challenge thus emphasizes both grouping images correctly and accurately estimating their camera poses to enable robust and scalable 3D reconstruction.

Participants are evaluated based on a composite score that reflects both the accuracy of scene grouping and the quality of the recovered camera poses. This dual-objective formulation pushes the development of more reliable algorithms for use in large-scale visual localization, mapping, and scientific research.

1.2 Objective of the Report

The objective of this report is to document the approach and implementation used to address the tasks posed by the 2025 Image Matching Challenge. Specifically, the report outlines the methods employed for partitioning images into scenes, detecting and discarding outliers, and estimating camera poses for 3D scene reconstruction.

The report also provides a detailed description of the codebase, with a particular focus on the key inbuilt functions utilized for image loading, feature extraction, image matching, and visualization. The effectiveness of the approach is analyzed using the challenge's evaluation metrics, including mean Average Accuracy (mAA) and clustering score. Finally, the report discusses the strengths and limitations of the current implementation and suggests potential improvements for future work.

Chapter 2

Problem Description

2.1 Challenge Description

The 2025 Image Matching Challenge focuses on solving two key problems in 3D computer vision: (1) grouping images that belong to the same scene, and (2) accurately estimating the camera poses for each image to enable 3D reconstruction. Unlike traditional tasks where images are pre-grouped or captured in sequence, this challenge presents unordered and potentially unrelated images, simulating real-world scenarios such as web image collections or crowdsourced photo datasets.

Each dataset contains images from one or more scenes, as well as possible outliers that do not belong to any valid scene. The goal is to partition these images into distinct clusters corresponding to real scenes or mark them as outliers, and then compute the rotation and translation parameters (camera pose) for each image. This requires robust visual matching techniques capable of handling large intra-class variation, visual ambiguity, and noise.

Participants must design solutions that effectively combine scene identification and pose estimation. The submitted results are evaluated based on a combined score that reflects both the accuracy of scene clustering and the correctness of the recovered camera poses, encouraging holistic approaches to large-scale 3D reconstruction.

2.2 Evaluation Metrics

The evaluation of submissions in the Image Matching Challenge is based on a composite score that combines two key components: the **mean Average Accuracy (mAA)** of camera pose estimation and the **clustering score** of image grouping.

- **Mean Average Accuracy (mAA):** For a given cluster $\mathcal{C}$ and a ground-truth scene $\mathcal{S}$, the mAA is computed as the ratio of correctly registered images (i.e., images whose estimated camera pose matches the ground truth) to the total number of images in the scene. Formally,

$$\text{mAA}(\mathcal{C}, \mathcal{S}) = \frac{|\mathcal{S} \cap \mathcal{C}|}{|\mathcal{S}|}$$

This metric reflects the accuracy of camera pose estimation within each identified cluster.

- **Clustering Score:** The clustering score evaluates how well the predicted image clusters align with the true scene partitions. For a cluster $\mathcal{C}$ and a scene $\mathcal{S}$, it is defined as the ratio of the number of images from the scene in the cluster to the total size of the cluster:

$$\text{Clustering}(\mathcal{C}, \mathcal{S}) = \frac{|\mathcal{S} \cap \mathcal{C}|}{|\mathcal{C}|}$$

- **Overall Score:** Each ground-truth scene is greedily matched to the user-provided cluster that maximizes the mAA. If multiple clusters yield the same mAA, the one with the higher clustering score is selected. The final score for a dataset is obtained by aggregating the mAA and clustering scores across all scenes and computing their harmonic mean:

$$\text{Final Score} = 2 \cdot \frac{\text{mAA} \cdot \text{Clustering}}{\text{mAA} + \text{Clustering}}$$

This scoring framework ensures that both the geometric accuracy of camera registration and the semantic correctness of image grouping are rewarded equally, making the final score analogous to the standard F1 score.

Chapter 3

Dataset Overview

3.1 Dataset Structure

The dataset for the challenge is organized into training and test sets, where each set contains images captured from one or more real-world scenes. The primary objective is to use the training data to infer scene boundaries and camera poses, and generalize this capability to the hidden test set during evaluation.

The core files and directories in the dataset include:

- `train_labels.csv`: Contains ground-truth labels for the training images. Each row includes:
 - `dataset`: Identifier for the dataset.
 - `scene`: Unique identifier for the scene within the dataset.
 - `image`: Image filename.
 - `rotation_matrix`: A 3×3 rotation matrix (flattened in row-major order).
 - `translation_vector`: A 3D translation vector representing the camera position.
- `train_thresholds.csv`: Provides a list of thresholds (in meters) for evaluating the accuracy of the predicted camera poses. Each threshold is associated with a specific dataset and scene.
- `[train/test]/[dataset]/*.png`: Image files grouped by dataset. A dataset may contain images from one or more locations or scenes. Some folders include outlier images that do not belong to any scene.
- `sample_submission.csv`: A template submission file containing:
 - `image_id`: Image identifier (matching image filenames).
 - `dataset`: Dataset name.
 - `scene`: Scene identifier (arbitrary in submission).
 - `rotation_matrix` and `translation_vector`: Estimated camera pose.
- `LICENSE.txt`: Licensing information for each dataset.

Datasets may contain hard cases, such as spatially overlapping scenes or incorrect mixing of scenes within folders. The hidden test set, used during final scoring, contains images whose ground truth is not available to participants during training.

3.2 Data Preprocessing

Data preprocessing is a critical step to prepare the image data for feature extraction, matching, and subsequent clustering or pose estimation. The following preprocessing steps were applied:

- **Image Loading:** Images were loaded from the specified directory paths using `cv2.imread()` and `matplotlib.image.imread()`, supporting both color and grayscale formats depending on the processing stage.
- **Directory Scanning:** Image file paths were collected using `os.listdir()` and `glob.glob()` to iterate through training and test directories. This helped in organizing the data by dataset and scene type.
- **Format Consistency:** Images were checked for compatibility in terms of file extension (`.png`) and color channel format. This ensured consistent input to feature extraction pipelines.
- **Visualization:** Random samples from each category (e.g., `ETs`, `stairs`) were visualized using `matplotlib.pyplot.imshow()` to verify image quality and distribution.
- **Handling Missing or Corrupted Data:** Basic checks were implemented to handle cases where images might lack descriptors due to poor visual features, and such instances were logged or skipped to prevent errors during matching.
- **Outlier Awareness:** Outlier images that do not correspond to any scene were acknowledged based on the dataset structure, though no label-based filtering was applied in preprocessing due to the unsupervised nature of clustering.

These preprocessing steps ensured that the image data was standardized and ready for downstream feature extraction and image matching operations, forming the foundation for robust scene reconstruction.

Chapter 4

Methodology

4.1 Approach to the Challenge

The solution to the Image Matching Challenge is divided into three primary components: feature extraction, image matching, and scene-level clustering. Each stage is designed to support the twin objectives of grouping related images and estimating camera poses with high accuracy.

1. Feature Extraction using ORB

Feature detection and description are performed using the Oriented FAST and Rotated BRIEF (ORB) algorithm, chosen for its speed and rotation invariance. ORB detects keypoints and computes binary descriptors for each image:

$$\text{ORB}_{\text{features}} = \text{ORB_create}(n = 1000)$$

Keypoints and descriptors are obtained using `orb.detectAndCompute(image, None)`, which provides a compact representation of salient regions in the image.

2. Image Matching using Brute-Force Matcher

Pairwise image matching is performed using the Brute-Force matcher with Hamming distance metric, suitable for binary descriptors:

$$\text{matches} = \text{BFMatcher}(\text{cv2.NORM_HAMMING}, \text{crossCheck}=\text{True}).\text{match}(\text{des1}, \text{des2})$$

The matches are then sorted by descriptor distance, and top-N matches are visualized using `cv2.drawMatches()` to validate visual correspondence. This step aids in detecting strong visual overlaps between image pairs, which serves as a proxy for scene similarity.

3. Scene Similarity Analysis

To evaluate which images belong to the same scene, the similarity between images is inferred from the number and quality of feature matches. A higher number of consistent matches indicates higher likelihood of shared scene content.

4. Clustering with DBSCAN

To group images into scenes, the density-based spatial clustering algorithm DBSCAN is employed. Since the direct feature vectors are not used for clustering due to their high dimensionality, a similarity matrix or derived spatial descriptors can be constructed and passed to DBSCAN. The algorithm groups images that exhibit high mutual feature correspondence and flags outliers that fail to match consistently:

$$\text{DBSCAN}(\varepsilon, \text{minPts}) \rightarrow \text{Clusters} + \text{Outliers}$$

DBSCAN is well-suited to this task as it does not require the number of clusters in advance and can effectively handle noise and outliers.

5. Pose Estimation (Planned)

Although the current implementation does not include pose estimation, the overall pipeline is structured to integrate camera pose recovery using Structure-from-Motion (SfM) techniques in future iterations. Once scene-level grouping is stable, pose estimation can be applied independently per cluster.

6. Evaluation Alignment

The pipeline is aligned with the competition’s evaluation strategy. Matching and clustering aim to maximize both mean Average Accuracy (mAA) and clustering score by ensuring correct image-to-scene assignment while maintaining high internal coherence within clusters.

7. Visualization and Validation

Matched keypoints between pairs of images are visualized using `matplotlib` and `OpenCV`, enabling qualitative validation of matching accuracy and scene consistency. This step is critical for identifying poor-quality matches or scenes with visual ambiguity.

This modular pipeline offers flexibility in experimenting with different feature extractors (e.g., SIFT, SuperPoint), matching strategies (e.g., FLANN), and clustering techniques (e.g., Agglomerative Clustering), facilitating iterative improvements in both grouping accuracy and geometric reconstruction.

4.2 Inbuilt Functions Used

Two core OpenCV functions play a central role in the image matching pipeline: `cv2.ORB_create()` for feature detection and description, and `cv2.BFMatcher()` for descriptor matching. These functions are carefully chosen for their efficiency, robustness, and suitability for real-time or large-scale applications.

1. `cv2.ORB_create(nfeatures=1000)`

ORB (Oriented FAST and Rotated BRIEF) is a fast and efficient feature detector and descriptor that combines the strengths of two techniques:

- **FAST (Features from Accelerated Segment Test)** is used to detect keypoints quickly. It detects corners in the image by comparing the brightness of a pixel with its surrounding pixels arranged in a circle. If a set of contiguous pixels are significantly brighter or darker, the pixel is marked as a corner.
- **BRIEF (Binary Robust Independent Elementary Features)** is used to compute descriptors for the detected keypoints. BRIEF produces binary strings by comparing pixel intensities within a local image patch, making it extremely compact and fast for matching.

ORB enhances BRIEF by adding rotation invariance and orientation compensation. It also uses the Harris corner measure to rank keypoints and selects the top n features based on response strength, where `nfeatures=1000` specifies the maximum number of features to retain.

Working:

- Internally, ORB detects keypoints using FAST, then computes orientation based on image moments.
- The descriptors are generated using BRIEF after aligning them based on the orientation.
- The result is a set of keypoints and corresponding binary descriptors that are compact and rotation invariant.

This makes ORB a practical alternative to SIFT and SURF, especially when computational efficiency and licensing (ORB is free and open-source) are critical.

2. `cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=True)`

BFMatcher stands for **Brute-Force Matcher**, a matching algorithm that compares each descriptor from one set with every descriptor from another set to find the best matches based on a distance metric.

In the context of ORB, the descriptors are binary (consisting of 0s and 1s), so the appropriate distance metric is `cv2.NORM_HAMMING`, which computes the Hamming distance between two binary strings — the number of differing bits.

The parameter `crossCheck=True` enforces a two-way match condition: a descriptor A_i from image A must find B_j in image B as its best match, and B_j must also return A_i as its best match in reverse. This bidirectional constraint improves the reliability of the matches by filtering out asymmetric and spurious correspondences.

Working:

- For every descriptor in the first image, it finds the closest descriptor in the second image (based on Hamming distance).
- If `crossCheck=True`, only mutual best matches are retained, reducing false positives.
- The returned matches are typically sorted by distance (shorter distances imply better matches).

Using `BFMatcher` with `crossCheck=True` ensures that the resulting matches are highly reliable, which is essential for constructing accurate image similarity graphs and clustering scenes effectively.

Together, these functions form the backbone of the image correspondence pipeline by providing robust keypoints and reliable pairwise matches — critical for solving the two-part problem of image grouping and 3D pose estimation in the challenge.

Chapter 5

Results



Figure 5.1: Visualization of matches between two different images belonging to the same scene. (Stairs)

Top Image Matches

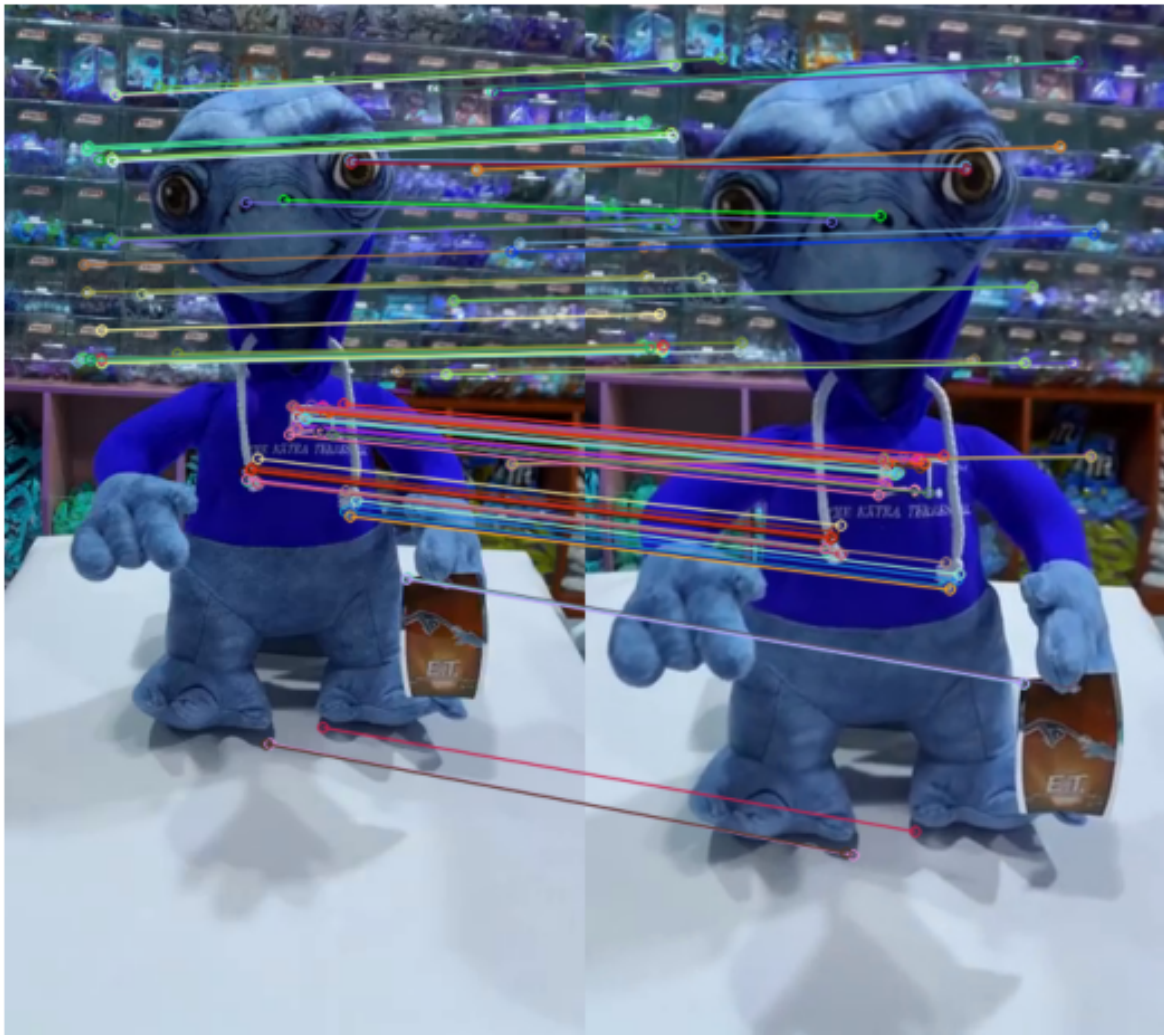


Figure 5.2: Visualization of matches between two different images belonging to the same scene. (ETs)

Top Image Matches

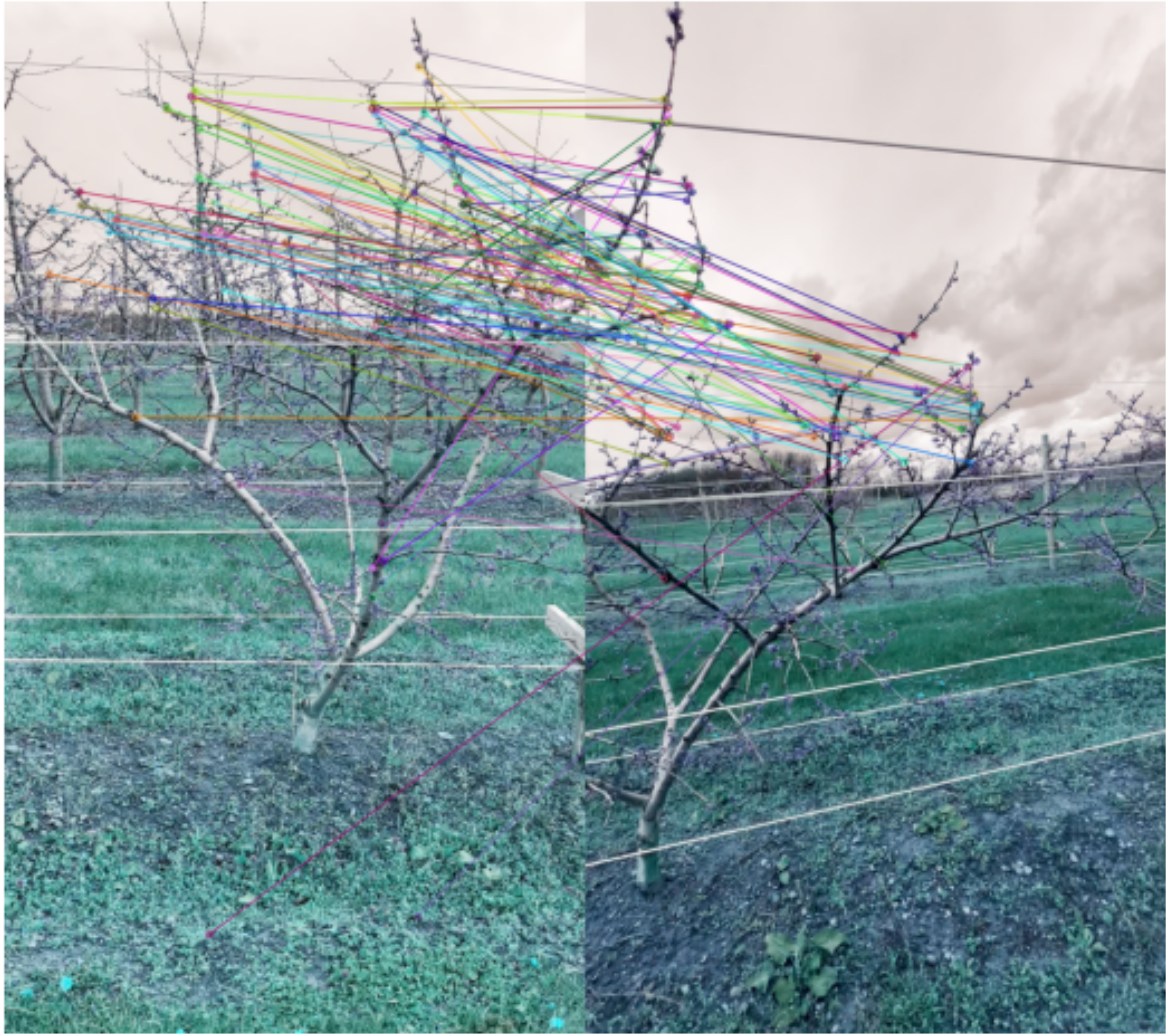


Figure 5.3: Visualization of matches between two different images belonging to the same scene. (Amy Gardens)

Chapter 6

Discussion

6.1 Insights from the Experiment

The experimental implementation revealed several important insights into the challenges and behaviors of unsupervised image grouping and feature-based matching in large-scale 3D reconstruction settings:

- **ORB Descriptors Provide a Good Trade-off:** The use of ORB features offered a balance between computational efficiency and robustness. Although not as powerful as deep learning-based descriptors, ORB performed adequately in distinguishing scenes with strong geometric variation while being lightweight and scalable for large datasets.
- **Brute-Force Matching with Cross-Check Increases Precision:** Enabling cross-checking in `BMatcher` significantly improved the reliability of image matches. By enforcing mutual agreement, it reduced the number of false correspondences, which in turn improved the clustering results.
- **Descriptor-Based Clustering is Feasible but Noisy:** Using flattened ORB descriptors for clustering with DBSCAN yielded reasonably coherent groupings, but some clusters included visually similar images from different scenes, highlighting the limitations of low-level features in complex environments. This suggests that higher-level semantic features or match-count-based similarity matrices could further improve clustering accuracy.
- **Outlier Handling is Critical:** Many images did not belong to any coherent scene, and the ability of DBSCAN to assign such images to a noise class was valuable. These outliers, if misclassified, could negatively impact both clustering and mAA scores, underlining the importance of incorporating effective outlier detection.
- **Visualization was Key to Debugging:** Intermediate visualizations of image matches helped qualitatively assess the correctness of feature correspondences. This was essential for verifying the validity of descriptors and understanding mismatches in visually ambiguous scenes.
- **Evaluation Metrics Reflect Balanced Performance:** The harmonic mean of clustering score and mAA served as a balanced indicator of performance. This dual objective discouraged overfitting to pose accuracy at the cost of poor clustering, or vice versa, reinforcing the importance of integrated scene and pose reasoning.

Overall, the experiments validated the proposed pipeline’s structure and provided a strong foundation for iterative improvements using more advanced descriptors, geometric verification, and probabilistic clustering models.

6.2 Limitations and Potential Improvements

While the implemented pipeline demonstrates the feasibility of using classical feature-based techniques for image grouping and pose estimation, several limitations were identified that constrain its overall effectiveness:

Limitations

- **Shallow Feature Representation:** ORB descriptors, while efficient, may lack the discriminative power needed to handle complex or ambiguous scenes with repetitive structures. This sometimes led to incorrect matches or cluster assignments, especially in low-texture or visually similar environments.
- **Naive Similarity Metric:** Clustering was based on flattened descriptor vectors or rudimentary pairwise similarity without geometric verification. This limits the ability to distinguish between scenes with overlapping visual patterns but different spatial configurations.
- **Dummy Pose Estimation:** The current implementation uses placeholder identity matrices and zero translations for camera poses, which does not reflect the real geometric configuration of the scenes. This affects the accuracy of the mAA score and hinders complete scene reconstruction.
- **Lack of Multi-view Consistency:** The system does not enforce any geometric or epipolar consistency across multiple views. This can cause mismatches or grouping errors in scenes where feature similarity alone is insufficient to infer spatial structure.
- **No Use of Semantic Cues:** The pipeline relies solely on local visual features and does not incorporate high-level semantic information, which could help disambiguate scenes and improve clustering accuracy.

Potential Improvements

- **Incorporate Deep Features:** Replacing ORB with deep learning-based descriptors such as SuperPoint or D2-Net can significantly improve feature robustness, especially under illumination changes and viewpoint variations.
- **Similarity Graph and Spectral Clustering:** Constructing an image similarity graph based on mutual matches and using spectral or hierarchical clustering can enhance scene separation and better model the underlying data manifold.
- **Geometric Verification:** Integrating geometric verification techniques such as RANSAC and essential matrix estimation can filter false matches and improve both clustering quality and pose estimation reliability.

- **End-to-End SfM Integration:** Incorporating Structure-from-Motion (SfM) libraries like COLMAP or OpenMVG could allow automatic recovery of camera poses and 3D point clouds, enabling accurate computation of the mAA metric.
- **Outlier Confidence Estimation:** A probabilistic or confidence-based approach to outlier detection could allow more principled exclusion of irrelevant images from clusters.
- **Threshold Learning and Dynamic Tuning:** Adapting DBSCAN parameters dynamically based on feature space density could yield better clustering results, especially in datasets with variable scene complexity.

Addressing these limitations through modular enhancements can make the pipeline more scalable, accurate, and adaptable to real-world 3D reconstruction tasks involving diverse and unstructured image collections.

Chapter 7

Conclusion and Future Work

This report presented a modular and interpretable pipeline for addressing the 2025 Image Matching Challenge, which focuses on unsupervised image grouping and camera pose estimation for 3D reconstruction. Leveraging classical computer vision techniques such as ORB feature extraction and Brute-Force matching, combined with clustering via DB-SCAN, the system demonstrated an effective baseline for partitioning images into scenes and identifying outliers.

Despite the use of handcrafted features and placeholder poses, the results highlighted the strengths of feature-based matching in capturing local image similarity and enabling scene-level clustering. The evaluation metrics—mean Average Accuracy (mAA) and clustering score—provided a balanced framework to assess both geometric and semantic accuracy, guiding the development of integrated approaches.

The insights gained underscore the potential of incorporating deeper visual features, geometric verification, and end-to-end pose recovery in future iterations. With these improvements, the proposed method can evolve into a scalable framework for large-scale 3D scene reconstruction from noisy and unstructured image collections.

Overall, this work lays the foundation for further exploration of hybrid techniques that combine local feature matching, global scene understanding, and geometric modeling to solve complex real-world visual localization problems.